

The elephant in the room

The discussion I want to open up for debate this month is about Big Data – the peg for our cover story this time. If you're completely new to this concept let me try to capture what it is in as few words as possible.

Many definitions have been written, but the best way to grasp the concept is by thinking about its enormity. Chew on this fact: every day we create 2.5 quintillion bytes of data (that's 18 zeros). Where does this data come from? It comes from every little status update you make, every bank transaction you complete, and even from the chatter of the (so far) lifeless network of embedded systems enveloping the world.

It's not the data itself, but the way corporations and governments have learned to mine this data that's interesting. Using sophisticated algorithms they can connect the dots in ways you wouldn't have imagined possible. Every aspect of our digital existence is on the brink of a profound transformation thanks to Big Data.

For example, with the proliferation of sensors, Big Data analysis will be able to predict the occurrence of natural disasters in the future. Imagine you're leaving home and your phone beeps "There's an 80 per cent chance of cloudburst over Mumbai. Either stay indoors or carry a life raft!" And that's the most pedestrian example I can think of.

Imagine a scenario in which data streams from centralised debt-risk lists, location sensors and credit card purchases are all being crunched by a powerful supercomputer. A man who's severely in debt is located via GPS in a shady part of town. Next, he pays for ammunition for his licenced rifle using his almost maxed out credit card. Somewhere a red ball drops, "72 per cent chance of criminal activity, John Anderton!" The shameless Minority Report reference aside, this is what the future holds, and we won't need three emaciated psychics in a protein soup to achieve this; just Big Data – matching patterns in creative ways from seemingly abstract and random 1s and 0s. What society does once it knows those outcomes is where those prickly moral questions come into the picture. Would we arrest people preemptively?

Let's leave the distant and perhaps scary future aside for now. Big Data will soon be used positively to create tailor-made experiences in fields such as education. Courses designed

to suit individual needs – be it pace, teaching style or difficulty – will help students learn more efficiently. An emerging science called "Social Physics" being evangelised by a group of MIT researchers aims to make sense of all the digital bread crumbs we leave behind in our daily life to form "patterns of behaviour". They analysed digital data in real settings over a long period of time, and were able to discover conclusively that people change how they behave in measurable ways when they fall ill. By tracking mobility and call patterns, the researchers were able to accurately tell when a subject was coming down with flu, even before the subject began to feel unwell.

Today, targeted ads are already helping you find products you need even before you realise you need them – for example, ads will suggest a protective case when you post a status update about a new smartphone purchase. If you're getting a little queasy at the prospect of so much being known about you, wait till you read our cover story! Our premise, in fact, is that you're being packaged and sold as a product by the very same entities whose platforms you're so recklessly willing to use for free. There are tons of examples in the cover story of misuse of Big Data, such as a convenience store accidentally revealing a girl's pregnancy to her family based on analysis of her purchasing pattern, even before she knew it herself!

You might be wondering, in the light of all this negative information, what options you have as consumers. Sadly the question of whether to participate or not in this ocean of big data is no longer a choice. Going off the grid is only a fanciful notion. It's not something you can actually afford to do; not if you want to be a part of normal society. "I'm not on Facebook or LinkedIn," is increasingly not going to be an acceptable answer. This is where I'd like to bring your attention to the opening para of this editorial. Help me settle the debate by writing to me: Is Big Data a nightmare in the waiting or is it the ticket to a better life for all mankind? 📧



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"Sadly the question of whether to participate or not in this ocean of big data is no longer a choice."

A handwritten signature in black ink, appearing to read 'Siddharth Parwatay'.

Got feedback about this column?
Drop me a line at:
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